QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

seedsigner @ ada81cde050e · SeedSigner — public repository (passive review)

Organization	QSI
Report date	2026-10-29 05:36 UTC
Report version	1
Audit run	54ad7f7a-4171-4408-aacb-3479d20b9be8 · Document audit · 2026-10-29 05:36 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`effe8cd836a4aeb1f5d672cd26315f8ade845f2d3a8f3b60aa0ba39a6416ff83`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

1	0	0	1	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	0
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	1
Improvement recommendation	0
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-06413	Medium	Quantum-vulnerable ECDSA: transaction and message signing (1 file(s)) confirmed by the analyst	41.8

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Analyst review: 1 confirmed, 0 not applicable, 0 dismissed as false positives.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/SeedSigner/seedsigner @ ada81cde050e46c19cb6e3fa73872b73955a138c — evidence links open the lines at this commit.

Supplied inputs

74 files, 986658 bytes in total, 0 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
dependencies	checked	4	pyproject.toml: 0 dependencies parsed (0 pinned); requirements-raspi.txt: 4 dependencies parsed (4 pinned); requirements.txt: 6 dependencies parsed (1 pinned)
source_code	checked	70	setup.py: 8 lines scanned with 16 rule patterns and 123 registry patterns; src/main.py: 53 lines scanned with 16 rule patterns and 123 registry patterns; src/seedsigner/__init__.py: 0 lines scanned with 16 rule patterns and 123 registry patterns
osv_advisories	checked	1	5 pinned packages queried at OSV.dev; 0 affected

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 75 (input files: C)

Architecture overview

Static review of the source tree seedsigner at ada81cde050e46c19cb6e3fa73872b73955a138c. 74 files analysed; of 166 files, 57 are test, fuzzing, benchmark or example code and were left out, 34 are not read by any analyzer, 1 exceed the per-file limit and 0 did not fit the total limit. 3 files come from directories a .gitmodules pins and are read as shipped source rather than skipped as a dependency cache. No requests were sent to the owner's systems.

Overall security posture

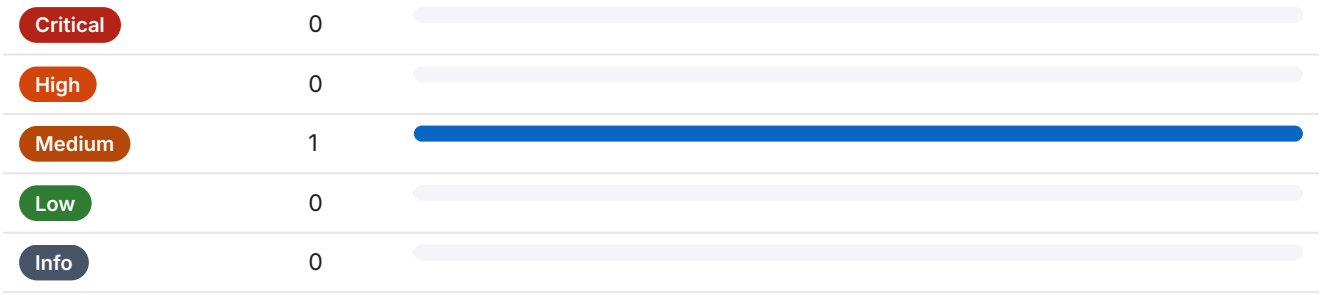
100
Indicative posture index (0–100)

14.3%
Quantum-vulnerable share of inventory

7
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-06413	Quantum-vulnerable ECDSA: transaction and message signing (1 file(s))	Long-term post-quantum risk	Medium	Medium

Detailed findings

QSI-2026-06413 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-29): confirmed by the analyst.** Correct, and it is the only row of its kind because it is the only place SeedSigner touches the primitive itself: src/seedsigner/helpers/embit_utils.py:10, the wrapper around embit. Everything that signs is embit's, and embit is not in this tree. Three consequences worth stating rather than leaving implied. First, the honest reading of "1 Shor-breakable row" is "this repository delegates its signing", not "this device is nearly post-quantum". Second, the pin is exact and hash-locked (embit==0.8.0 with a sha256 --hash), so the audited exposure is a specific, reproducible artefact rather than a version range — which is what makes the delegation auditable at all, and is the opposite of a criticism. Third, the migration question for SeedSigner is therefore not "when will SeedSigner adopt a post-quantum signature" but "when will embit", and after that, "how fast can a hash-locked pin be moved" — which is a release away, not a redesign. Compare Krux, which pins the same library as a git submodule and so carries the same code inside its own tree: 98 rows and 64 Shor-breakable for the same cryptography, differing only in where it is kept.

3 reference(s) to ECDSA across 1 file(s), e.g. src/seedsigner/helpers/embit_utils.py:10. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	src/seedsigner/helpers/embit_utils.py
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:36 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/seedsigner/helpers/embit_utils.py:10 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · src/seedsigner/helpers/embit_utils.py · line 10 · SHA-256 1e5604518dfe3ecc… · File c087d19323033176…

```
 9 | from embit.networks import NETWORKS
10 | from embit.util import secp256k1
11 |
```

file_excerpt · src/seedsigner/helpers/embit_utils.py · line 204 · SHA-256 1060384364a8c023… · File c087d19323033176…

```
203 |     prv = root.derive(derivation).key
204 |     sig = secp256k1.ecdsa_sign_recoverable(msghash, prv._secret)
205 |     flag = sig[64]
```

file_excerpt · src/seedsigner/helpers/embit_utils.py · line 209 · SHA-256 1e4a07630105e8ba… · File c087d19323033176…

```
208 |     flag = bytes([27 + flag + c])
209 |     ser = flag + secp256k1.ecdsa_signature_serialize_compact(sig._sig)
210 |     return b2a_base64(ser).strip().decode()
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

Evidence

ID	Type	Data source	SHA-256
QSI-2026-06413	file_excerpt	src/seedsigner/helpers/embit_utils.py (line 10)	1e5604518dfe3ecc40df631b9df1ab8a9e5e838974ac1c814502658edc7d3fd3
QSI-2026-06413	file_excerpt	src/seedsigner/helpers/embit_utils.py (line 204)	1060384364a8c023d525064551aeed7ece6fd577f80394a1a55de0cce5980c19
QSI-2026-06413	file_excerpt	src/seedsigner/helpers/embit_utils.py (line 209)	1e4a07630105e8ba280ad6a4b597896533871c3bfe4f6dc7bdfcc27b6efb125e

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	ECDSA (1)	QSI-2026-06413	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
reduced margin (Grover), approved	2
Grover: acceptable to NIST, below ECCG/ANSSI guidance	4
vulnerable (Shor)	1

Detailed findings: QSI-2026-06413

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDSA (1)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	PBKDF2 (1), SHA-2 (5)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (5)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-06413 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06413 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
None			

60 days

Priority	ID	Action	Type
None			

90 days

Priority	ID	Action	Type
P3	QSI-2026-06413	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-06413	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-06413	Re-run the inventory after refactoring.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

1. A. Chailloux, M. Naya-Plasencia, A. Schrottenloher — An Efficient Quantum Collision Search Algorithm and Implications on Symmetric Cryptography (ASIACRYPT 2017; IACR ePrint 2017/847) — IACR ePrint. https://eprint.iacr.org/2017/847 (peer-reviewed article: Õ(2^{2n/5}) time, O(n) qubits, Õ(2^{n/5}) classical memory)
2. C. Chevignard, P.-A. Fouque, A. Schrottenloher — Reducing the Number of Qubits in Quantum Discrete Logarithms on Elliptic Curves (IACR ePrint 2026/280) — IACR ePrint. https://eprint.iacr.org/2026/280 (revision of a EUROCRYPT 2026 paper: 3.12n + o(n) qubits; for n = 256, 1193 qubits and 22 runs of 2^38.98 Toffoli gates)
3. C. Zalka — Grover's quantum searching algorithm is optimal (Phys. Rev. A 60, 2746, 1999; arXiv:quant-ph/9711070) — APS. https://doi.org/10.1103/PhysRevA.60.2746 (peer-reviewed article: optimal for any success probability; parallel search gains no more than splitting the space)
4. Common Vulnerability Scoring System v3.1: Specification Document — FIRST. https://www.first.org/cvss/v3.1/specification-document (final)
5. D. J. Bernstein — Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete? (2009) — cr.yp.to. https://cr.yp.to/hash/collisioncost-20090823.pdf (paper dated 2009-08-23)
6. D. Litinski — How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates (arXiv:2306.08585) — arXiv. https://arxiv.org/abs/2306.08585 (preprint (2023-06-14): 6000 modules × 1152 physical qubits, one key per 10 minutes under its architecture assumptions)
7. FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM) — NIST. https://csrc.nist.gov/pubs/fips/203/final (final (August 2024))
8. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
9. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
10. G. Brassard, P. Høyer, A. Tapp — Quantum Algorithm for the Collision Problem (LATIN '98; arXiv:quant-ph/9705002) — Springer / arXiv. https://arxiv.org/abs/quant-ph/9705002 (peer-reviewed article: collisions of r-to-one functions in O((N/r)^{1/3}) expected evaluations)
11. J. Proos, C. Zalka — Shor's discrete logarithm quantum algorithm for elliptic curves (Quantum Inf. Comput. 3(4), 2003; arXiv:quant-ph/0301141) — Quantum Information & Computation. https://doi.org/10.26421/QIC3.4-3 (peer-reviewed article)
12. L. K. Grover — A fast quantum mechanical algorithm for database search (STOC 1996) — ACM. https://doi.org/10.1145/237814.237866 (peer-reviewed article)
13. M. Amy et al. — Estimating the cost of generic quantum pre-image attacks on SHA-2 and SHA-3 (SAC 2016; arXiv:1603.09383) — Springer / arXiv. https://arxiv.org/abs/1603.09383 (peer-reviewed article: SHA-256 about 2^153.8 surface-code cycles on 2^12.6 logical qubits; SHA3-256 about 2^146.5 cycles on 2^20)
14. M. Cain et al. — Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits (arXiv:2603.28627) — arXiv. https://arxiv.org/abs/2603.28627 (preprint (2026-03-30): P-256 in a few days with 26,000 physical neutral-atom qubits; RSA-2048 one to two orders of magnitude longer)
15. M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018) — IEEE. https://doi.org/10.1109/MSP.2018.3761723 (peer-reviewed article; source of the x + y > z migration inequality)
16. M. Roetteler, M. Naehrig, K. Svore, K. Lauter — Quantum resource estimates for computing elliptic curve discrete logarithms (ASIACRYPT 2017) — IACR / arXiv. https://arxiv.org/abs/1706.06752 (peer-reviewed article: at most 9n + 2⌈log2 n⌉ + 10 qubits and 448 n^3 log2 n + 4090 n^3 Toffoli gates)
17. NIST IR 8547 — Transition to Post-Quantum Cryptography Standards — NIST. https://csrc.nist.gov/pubs/ir/8547/ipd (initial public draft (November 2024); proposes deprecating quantum-vulnerable public-key algorithms after 2030 and disallowing them after 2035 — dates may change in the final version)
18. NIST — Post-Quantum Cryptography Standardization: Evaluation Criteria, Security (Call for Proposals, Section 4.A.5) — NIST. https://csrc.nist.gov/Projects/Post-Quantum-Cryptography/Post-Quantum-Cryptography-Standardization/Evaluation-Criteria/Security-(Evaluation-Criteria) (security categories: AES-128 key search 2^170/MAXDEPTH quantum gates, AES-192 2^233/MAXDEPTH, AES-256 2^298/MAXDEPTH; MAXDEPTH 2^40–2^96)
19. P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997) — SIAM. https://doi.org/10.1137/S0097539795293172 (peer-reviewed article)
20. R. Babbush et al. — Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (arXiv:2603.28846) — arXiv. https://arxiv.org/abs/2603.28846 (preprint (2026-03-30, v2 2026-04-15): 256-bit ECDLP with < 1200 logical qubits and < 90 M Toffoli gates (or < 1450 and < 70 M); < 500,000 physical qubits at 10^-3 error, minutes)

21. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Location	Quantum status	Classical status
PBKDF2		src/seedsigner/models/seed.py:186	reduced margin (Grover), approved	acceptable
SHA-2	SHA-512	src/seedsigner/models/seed.py:181	reduced margin (Grover), approved	acceptable
SHA-2	SHA-256	src/seedsigner/gui/screens/tools_screens.py:112	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	src/seedsigner/helpers/mnemonic_generation.py:74	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	src/seedsigner/helpers/ur2/xoshiro256.py:66	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	src/seedsigner/views/tools_views.py:183	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
ECDSA	secp256k1	src/seedsigner/helpers/embit_utils.py:10	vulnerable (Shor)	acceptable

B. JSON

qsi.report/1 · SHA-256 effe8cd836a4aeb1f5d672cd26315f8ade845f2d3a8f3b60aa0ba39a6416ff83

C. Input files

File	Size	SHA-256	Rec
pyproject.toml	2006	`806812aaa643b69646850af44736db858192095ae1da0dd6432fa9a52a2a9331`	0
requirements-raspi.txt	98	`97a87553c8c5e02b39450fd57707fef2d68d1564c89402d3d25be468eb3c8481`	0
requirements.txt	7698	`f43fdf89d5cfe76e4f1d01d27df6c5030d6aa1767217d9d8c855e4a5fe34b45a`	0
setup.py	181	`d79d8ecf1aa4f4e496d725ea43fa2082d91016d760ad1581aae0ca8e5be61c4b`	0
src/main.py	1540	`0fec0abc13bf4f8d754e78ee16649b51028a85c3286f2abcb860fcc0b723bf27`	0
src/seedsigner/__init__.py	0	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`	0
src/seedsigner/controller.py	18529	`2b536116eb906cb6b30ef1ee0aeb85ff3b6e5dd4cb49f30ce9cf55ede4cbb97d`	0
src/seedsigner/gui/__init__.py	30	`dbcabb7d019bd42ebf2861fff87f66cbd004a652b0f5ea912505bc3b4dbe3258`	0
src/seedsigner/gui/components.py	79525	`9bf81d18f771279eaf856a536a59a54d9b445fb7fe777ce1a0dbf412e844af93`	0
src/seedsigner/gui/keyboard.py	25548	`dbfa96f48aec0b0d6cf7d83f4b493aca81fa4f67fee4d7c11e310c622e393334`	0
src/seedsigner/gui/renderer.py	4501	`308c77a3e190df410245c7c2762e2ee7464fa73933d6da4e8acc2a936460af67`	0
src/seedsigner/gui/screens/__init__.py	21	`a5496c911fdee0f9907e2d37723907443f11db5d81eb9559a8a527aba51a4b9f`	0

src/seedsigner/gui/screens/psbt_screens.py	33501	d8fae1241725686353d884bf816325bcff936d64ad831700234ecc943354063b	0
src/seedsigner/gui/screens/scan_screens.py	13062	938d846d7c518a035ab0b757fd87ee2c75cbe64b00ed5959e42592f35d460696	0
src/seedsigner/gui/screens/screen.py	55000	0212127e2533cdc72d421fb0ebc58f8f3278f1d3a4960ba9941abb01051f8eff	0
src/seedsigner/gui/screens/seed_screens.py	70769	c0b9eca6e4c76947997ba1352d3971ac259391f4746b19d74ba24d89bf85b17a	0
src/seedsigner/gui/screens/settings_screens.py	19504	e379674269d2edcce5f794a41b05896f4b692db39d5e19e3ea774415232c9a49	0
src/seedsigner/gui/screens/tools_screens.py	31245	4e407f8bfd50fe12fcd6e32037b44895abea4c2d196df6bdcaf228839bd03da4	0
src/seedsigner/gui/toast.py	13196	ec4cb0206b3d3d9903be66dfedee0bf920ade3bb71fb1f8496de8988bab7b245	0
src/seedsigner/hardware/__init__.py	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855	0
src/seedsigner/hardware/buttons.py	7931	4471efdfa2dfd502ab17f1a0e016bacb1031060293f9fb27c00d988748f625f7	0
src/seedsigner/hardware/camera.py	3241	4d1d61508ab5f83ebe23a20e2f9ea01fbd9269f3026865720d515497f717533b	0
src/seedsigner/hardware/displays/ST7789.py	4997	4d43e4974a43bb8aa12f7e0697ade09894a582085dfe67eec6b9582930479a7e	0
src/seedsigner/hardware/displays/__init__.py	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855	0
src/seedsigner/hardware/displays/display_driver.py	3068	99b99244032d27ce88865a1e4e5d3765e933a9b46a97e8f4ac63e4b05ca6a2d8	0
src/seedsigner/hardware/displays/ili9341.py	12574	c73eb77327bb882ecc78e7e5af355af969ce09d22435d981168cdab017fad998	0
src/seedsigner/hardware/displays/st7789_mpy.py	32972	71347193cdc17edb2aa54485f38befaa827dcadc02687d826d32f1597a3ce89f	0
src/seedsigner/hardware/microsd.py	2389	6b7dfe146c69b151eb4fe3dad3bd4e518875f04e30d51ec544d4d6432d317ac1	0
src/seedsigner/hardware/pivideostream.py	1819	0eb5e7d8333ed7d590d79eecdadf395d1c71d77772febea413dd1f924909e51c	0
src/seedsigner/helpers/__init__.py	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855	0
src/seedsigner/helpers/embit_utils.py	7727	c087d1932303317654685fd24b3753429964a5b4f63c5d927f1663bc9b18c3d1	0
src/seedsigner/helpers/l10n.py	162	611cbe564315817964999f3437eecc6a988e2b968a8169066b146e0d1f784686	0
src/seedsigner/helpers/mnemonic_generation.py	4630	1da00ac502a88ae77733681fd19f0fb61df03ebdee759b9ac78798fe9399c8fc	0
src/seedsigner/helpers/qr.py	4808	ed12c8c01002e52af9946e2c0b579c4fd692dee81612dabbc77e73fb1f0bdca8	0
src/seedsigner/helpers/ur2/__init__.py	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855	0
src/seedsigner/helpers/ur2/bytewords.py	4929	b652f89758fb84d2beafaeb01e77a8af5f3e7a201d5b3fa1473e01ef8418bfc1	0
src/seedsigner/helpers/ur2/cbor_lite.py	11208	313c06f8d23542fcd7eaa0c25ccb57d568571ba7afd6e630c5d9d418c2265913	0

src/seedsigner/helpers/ur2/constants.py	179	d7a4ccd6c624c69aec6983667399c6566dddb60134349e43443ac05b67956495	0
src/seedsigner/helpers/ur2/crc32.py	1037	888541118b69e5a3f587e1556f0787f6b644ff263811c6913843eb18b68513da	0
src/seedsigner/helpers/ur2/fountain_decoder.py	12114	4285d087cb3dcfb740c9c92329c4ba805cc3044f1a874fb0daa33405ef4fec3f	0
src/seedsigner/helpers/ur2/fountain_encoder.py	5244	155812c2f58fc22888b6b2ec601d40fe405f6a4cbc0be58794c8052bf411b25a	0
src/seedsigner/helpers/ur2/fountain_utils.py	1597	cbbc87708a1ad1f4e5494f04a4c29ca585946fa55fb0a4dd92711d7d40c20d42	0
src/seedsigner/helpers/ur2/random_sampler.py	1568	dcb23762815edd2a0b0fe48fba02fac5b3aba3d67c1f27bafe6823b1dda04b59	0
src/seedsigner/helpers/ur2/ur.py	494	b51393572a556ff4426412783b1902d1afc6241f26870ecf649eefb719880929	0
src/seedsigner/helpers/ur2/ur_decoder.py	5203	61fc0d5319f6915aab118bcd13e6aa25e0e021c5b4dcc6af2222ff30a14e9c6a	0
src/seedsigner/helpers/ur2/ur_encoder.py	2430	2a05145e5961347404ccf7230727eee85efa2822f0b39890ae42e6fe47202c1b	0
src/seedsigner/helpers/ur2/utils.py	1619	565724921749e1ed0e92b6beb38d4f00df2ec373b105928bed1f8c30da297c6d	0
src/seedsigner/helpers/ur2/xoshiro256.py	4397	d4d700a789502a4ba21f4a3a24a3c8381f13a3c7fb896b0e19dffc6948aef1d4	0
src/seedsigner/helpers/version.py	37956	70759ff9f528343912572b8ff4e1597cf08f4371029346e93b5875322ed3b7b7	0
src/seedsigner/models/__init__.py	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855	0
src/seedsigner/models/decode_qr.py	39285	ca35d03c9ede201893a01e5faceb96d58449e1600a28e564ae29d6a2b0248912	0
src/seedsigner/models/encode_qr.py	13142	cba66b93b19ed0831e27bec44c49e6e488cf1634f2b82e5c5f3e9cffe9695957	0
src/seedsigner/models/psbt_parser.py	82783	3f8cbba4d378aa4f82ac9a1d4c05b5b7017f9b720eb92c38d530f4aab3dafbe3	0
src/seedsigner/models/qr_type.py	930	56b4789b1bc55e2b129d0a1fe873f7b2ff2d7ddafe62f8fce6c0e4d3814b5230	0
src/seedsigner/models/seed.py	7768	4f612eeae59dd972bd9187818c6648a70efe7e258f4d5c81662f9175060a7658	0
src/seedsigner/models/seed_storage.py	2957	2c2ad94d462898ca0d83b7dc935750db96d60c173aa0422dffc41ca9b2cde686	0
src/seedsigner/models/settings.py	13398	864853cc67cd59d60c220bceb922c502fe06db3993de3de76acee11604e23c13	0
src/seedsigner/models/settings_definition.py	35542	ffdefda767a713a8c7a54dffd712739daa235b01efdffa2068fbaba86b241cce	0
src/seedsigner/models/singleton.py	1059	caaa41a60af389c8b5322d510536d310664fa4ef53e8957c9fab7f021c4e71ac	0
src/seedsigner/models/threads.py	1079	678a61d601a2c8a0d5ff3b1c110f781b07b17a2f152e20a6fcf55e95f6a531ee	0
src/seedsigner/resources/seedsigner-translations/.github/diff_report/diff_screenshots.py	6279	7a1ecba23853b008f72921dfb453b14959e35ed7d8f0ebfa906a4eb0908319f2	0

src/seedsigner/resources/seedsigner-translations/tools/extract_characters_from_babel_mo.py	2700	df7055b2f0c8460ee5fd9f1e995cb7a8346e43af628d90a2b08b46d7766cd7ff	0
src/seedsigner/resources/seedsigner-translations/tools/requirements.txt	6	68359c38ed3bf77f71bad8d21c219fbe8c094d44374b35d2974e202f8b24472b	0
src/seedsigner/views/__init__.py	50	07d90e6e3c9bc4127568b08e5bdf2725ac093964a987c50d52efe3e4dca8d505	0
src/seedsigner/views/psbt_views.py	41251	47fff120f41e2710226560d722b70d51a8bee0ece389b9b3567092988b413750	0
src/seedsigner/views/scan_views.py	9526	f49166b5417dbc649da3b2b7871308aae608f573312e105945de36568d3628d3	0
src/seedsigner/views/screensaver.py	9727	e3eb0f54536dcd7d02296c5e5c6aa34ffcae152e24e994cc67d1e4b626a9d3cc	0
src/seedsigner/views/seed_views.py	90475	4c977bac0dbf3e893b0e58b7fabdf0ab89b141f32717e038de659a097e470d4c	0
src/seedsigner/views/settings_views.py	15828	eaaa53d1250f6ba624cbddb0a696306f08187653a598f783002e51a5c3c09797	0
src/seedsigner/views/tools_views.py	34242	c9dde008d0e3565a27bb572529d6a9a23607fc9cfa1560e6d4b12ab68e11f537	0
src/seedsigner/views/view.py	16231	b92cb1e521001a270d678ef7d32126c7736fecd629f781251a860a4d349d957d	0
tools/mnemonic.py	6971	92d3059fc0eb86241d4d8839d6c64116423291526d093bb4f1b37fdd95c13f60	0
tools/seed_phrase_to_qr.py	1580	de71bf293c28d5c4e637c545a07685160c2aa92c9f64213d7a5259d9e91de66b	0
tools/write_versionfile.py	1602	2bb499c921b6f8cf04f40f835ca3a6cd92ea445c50deb893e90d38262e19f53a	0